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TOPIC1: Programming Question

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Recursion Programs (10)

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TOPIC2: JAVA Interview question

1. What is a function (method) in Java, and why is it used in programming?
2. What is the difference between a method with return type and a void method in Java?
3. What are method parameters, and how are arguments passed to a function in Java?
4. What is method overloading in Java, and how is it different from method overriding?
5. What is recursion in Java, and how does a recursive function work?
6. What are the advantages and disadvantages of using recursion in Java programs?
7. What is a base case in recursion, and why is it important?

8. What is dynamic programming, and how is it different from recursion?
9. What are memoization and tabulation in dynamic programming?
10. How can arrays be passed to functions in Java, and what happens when an array is modified inside a method?

TOPIC3: JAVA MCQ Question

1. What will be the output of this code?

```
public class Test {  
    static int add(int a, int b) {  
        return a + b;  
    }  
  
    public static void main(String[] args) {  
        System.out.println(add(5, 3));  
    }  
}
```

- A. 53
- B. 8
- C. 5
- D. Error

Answer: B. 8

2. What is the output of the recursive function?

```
static int fact(int n) {  
    if(n == 1) return 1;  
    return n * fact(n - 1);  
}
```

```
public static void main(String[] args) {  
    System.out.println(fact(4));  
}
```

- A. 24
- B. 12
- C. 16
- D. 4

Answer: A. 24

3. What will this code print?

```
static void update(int[] arr) {  
    arr[0] = 100;  
}
```

```
public static void main(String[] args) {  
    int[] arr = {1, 2, 3};  
    update(arr);  
    System.out.println(arr[0]);  
}
```

- A. 1
- B. 100
- C. 0
- D. Error

Answer: B. 100

4. What is the output?

```
static int sumArray(int[] arr) {  
    int sum = 0;
```

```
    for(int x : arr) sum += x;
    return sum;
}

public static void main(String[] args) {
    int[] arr = {1, 2, 3};
    System.out.println(sumArray(arr));
}
```

- A. 6
- B. 123
- C. 5
- D. Error

Answer: A. 6

5. What will this program print?

```
static void change(String s) {
    s = "Java";
}

public static void main(String[] args) {
    String s = "Python";
    change(s);
    System.out.println(s);
}
```

- A. Java
- B. Python
- C. Null
- D. Error

Answer: B. Python

6. What is the output of recursion?

```
static int sum(int n) {  
    if(n == 0) return 0;  
    return n + sum(n - 1);  
}  
  
public static void main(String[] args) {  
    System.out.println(sum(5));  
}
```

- A. 10
- B. 15
- C. 20
- D. 25

Answer: B. 15

7. What is the output?

```
static int fib(int n) {  
    if(n <= 1) return n;  
    return fib(n - 1) + fib(n - 2);  
}  
  
public static void main(String[] args) {  
    System.out.println(fib(5));  
}
```

- A. 3
- B. 5
- C. 8
- D. 13

Answer: B. 5

8. What will happen in this program?

```
static int dp(int n) {  
    if(n == 0) return 0;  
    if(n == 1) return 1;  
    return dp(n - 1) + dp(n - 2);  
}
```

- A. Dynamic Programming (optimized)
- B. Recursion without optimization
- C. Compilation error
- D. Infinite loop

Answer: B. Recursion without optimization

9. What is the output?

```
static String reverse(String s) {  
    return new StringBuilder(s).reverse().toString();  
}  
  
public static void main(String[] args) {  
    System.out.println(reverse("abc"));  
}
```

- A. abc
- B. cba
- C. bca
- D. Error

Answer: B. cba

10. What will be the output?

```
static int max(int[] arr) {  
    int m = arr[0];  
    for(int i : arr) {  
        if(i > m) m = i;  
    }  
    return m;  
}  
  
public static void main(String[] args) {  
    int[] arr = {3, 7, 2, 9};  
    System.out.println(max(arr));  
}
```

- A. 7
- B. 3
- C. 9
- D. 2

Answer: C. 9

TOPIC4: Aptitude Question

1. Basic Average

The average of 5 numbers is 20. If four of the numbers are 18, 22, 19, and 21, find the fifth number.

2. Consecutive Numbers

Find the average of first 10 natural numbers.

3. Change in Average

The average age of 6 students is 15 years. If one more student aged 21 joins them, what is the new average?

4. Weighted Average

A student scored 60, 70, and 80 marks in three subjects. Find the average marks.

5. Average Speed

A car travels 60 km at 30 km/h and another 60 km at 60 km/h. Find the average speed of the journey.

6. Missing Value

The average of 8 numbers is 25. If one number is removed, the average becomes 24. Find the removed number.

7. Combined Average

The average weight of 5 boys is 50 kg and that of 3 girls is 45 kg. Find the combined average weight.

8. Age Problem

The average age of a family of 4 members is 30 years. A new member is born. What is the new average age after 2 years?

9. Increase in Average

The average of 10 numbers is 40. If each number is increased by 5, what will be the new average?

10. Team Average

The average score of a cricket team of 11 players is 45 runs. If the captain scored 90 runs, find the average score of the remaining players.